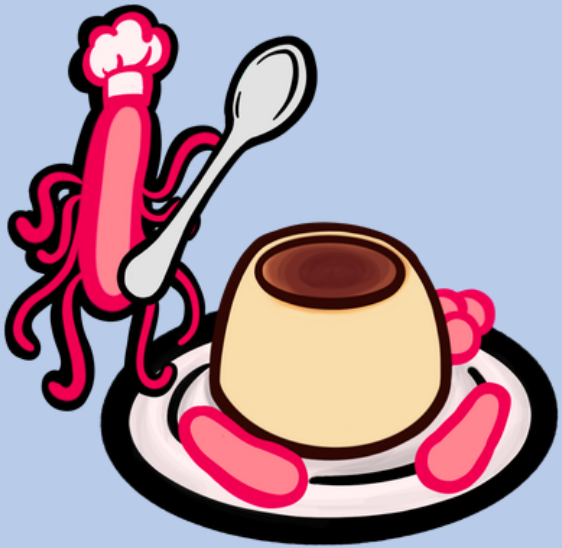
HUMAnN

Profile microbial gene and pathway abundance stratified by contributing species.



MetaPhlAn

Profile the presence and
abundance of microbial clades.



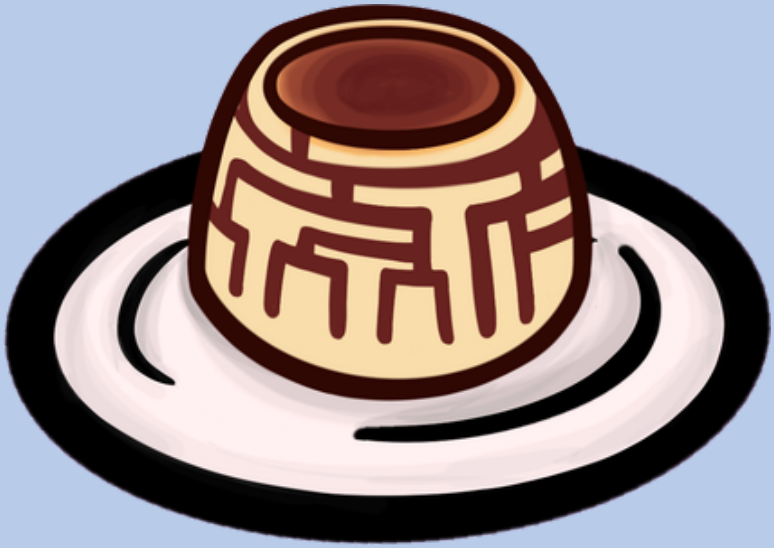
PICRUSt

Predict metagenome functional content from marker genes.



PhyloPhlAn

Reconstruct microbial phylogenetic trees and assign genome taxonomy.



MetaWIBELE

Identify and prioritize potentially bioactive gene families in microbial communities.



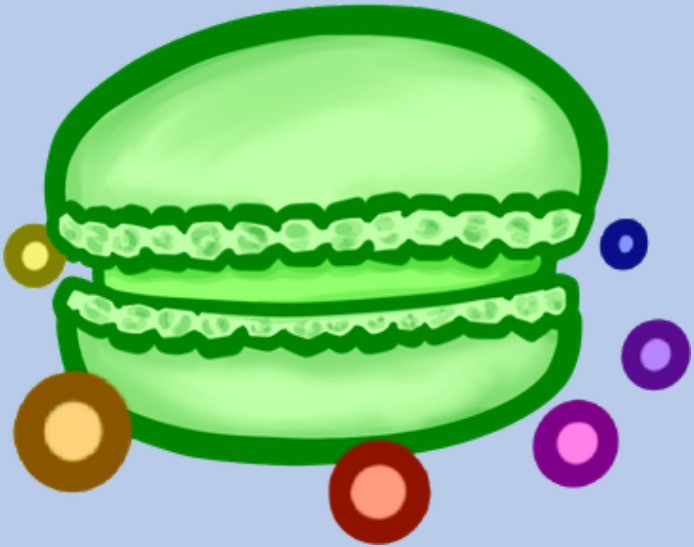
StrainPhlAn

Identify and track microbial strains based on single-nucleotide variants within marker genes.



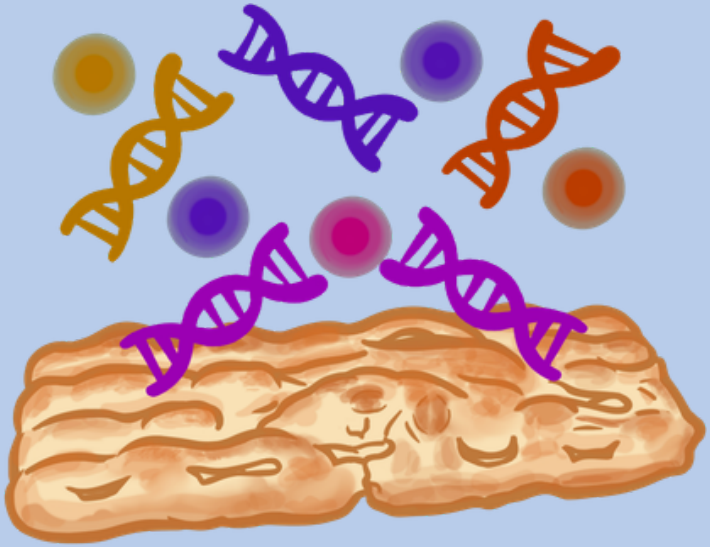
MACARRoN

Identify and prioritize potentially bioactive metabolites in microbial communities.



FUGAsseM

Predict functions for uncharacterized gene families based on microbial community data.



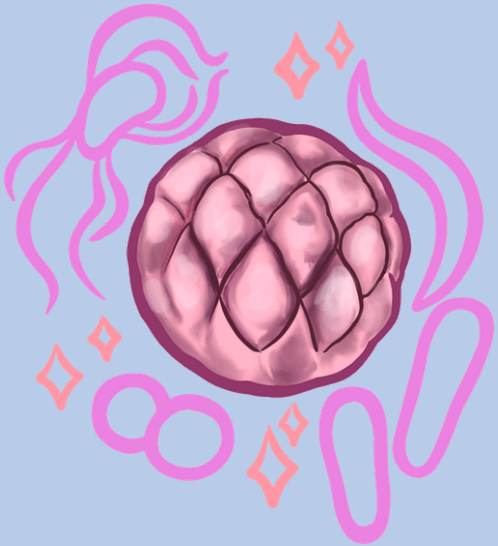
WAAFLE

Identify novel LGT events in assembled metagenomes.



MelonnPan

Predict metabolome content from microbial enzyme abundances.



ShortBRED

Profile proteins of interest
with very high specificity.



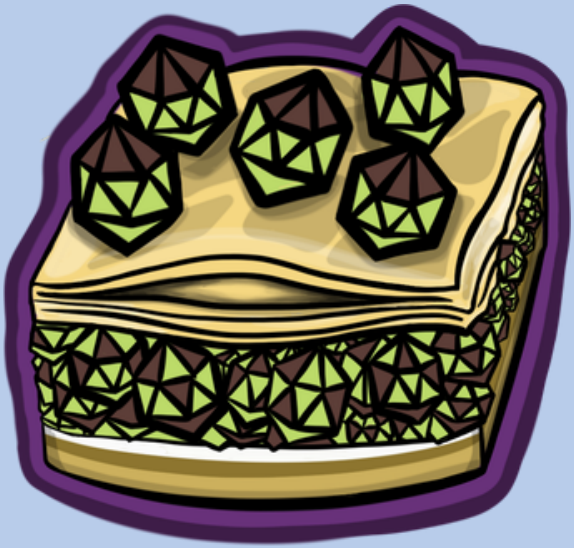
PPANINI

Prioritize microbial genes based
on their meta'omic properties.



BAQLaVa

Profile the presence and abundance
of known and novel viral taxa.



PARATHAA

Classify microbial amplicon sequences while preserving primer-specific taxonomic ambiguity.



PanPhlAn

Identify and track microbial strains
based on differential gene carriage.



MaAsLin

Associate arbitrarily complex
metadata with microbiome features.



MMUPHin

Correct batch effects and meta-analyze microbiome profiles across studies.



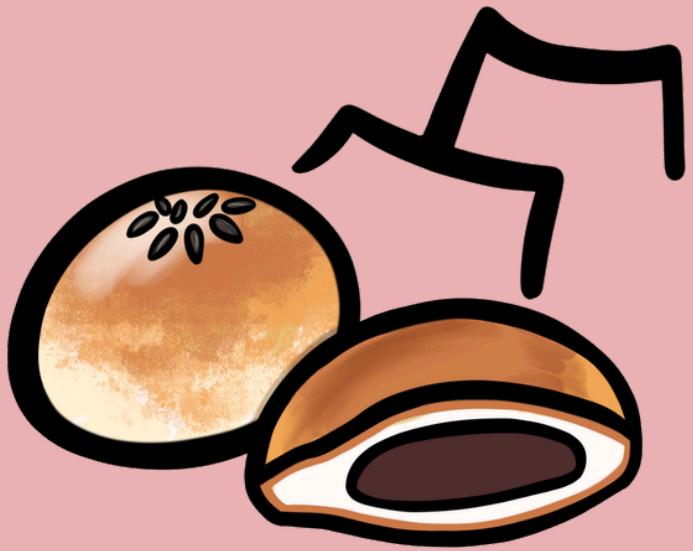
SparseDOSSA

Generate realistic microbial community abundance tables for statistical benchmarking.



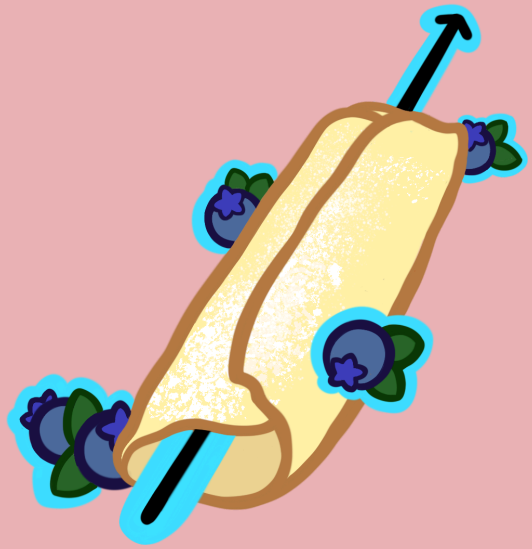
Anpan

Associate microbial strain properties and sample metadata by several methods.



CCREPE

Assess the statistical significance of similarity measures in compositional data.



HAllA

Perform well-powered comparisons
of paired, high-dimensional datasets.



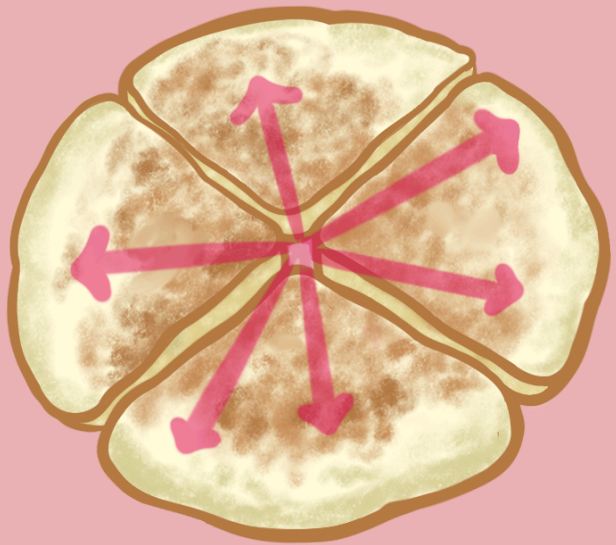
MicroPITA

Select samples for follow-up analysis in two-stage tiered studies.



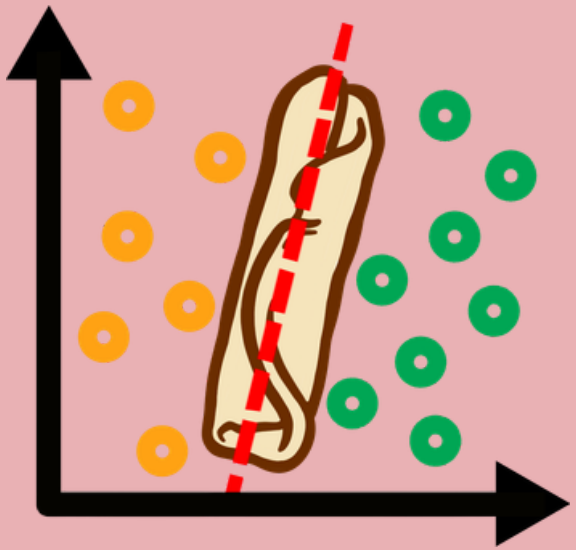
BAnOCC

Evaluate ecological similarity in compositional data using Bayesian methods.



LEfSe

Associate up to two metadata
with microbiome features.



ARepA

Extract and normalize ‘omics data from online repositories.



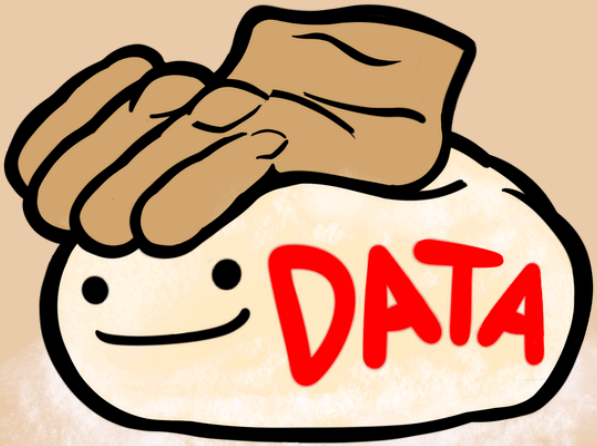
GraPhlAn

Generate cladograms and
decorate with metadata.



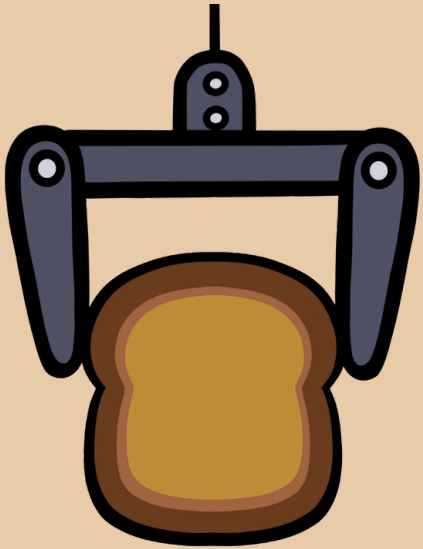
KneadData

Perform quality control of meta'omic sequencing reads, including host-read removal.



AnADAMA 2

Perform and document
automated scientific workflows.



bioBakery Workflows

A collection of workflows for executing common microbial community analyses.

